

PSP Oracle Exit - Audit Database Migration Tiptoe Runbook

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Goal

- Validate Deployment, App and Database Infrastructure built for Postgres Migration.
- Understand and learn the system behavior functional/non functional in the Postgres database.
- Validate 50% of P0 OLTP workflows and 35% of OLAP workflows.
- Validate the system under less load i.e. Weekend Traffic will have minimal impact.
- Validate Planned Rollback to prepare for any emergency Rollback.
- Validate all the above goals with the minimal customer and partner team impacts.

Tiptoe Summary

Status

Environment	Status
Production	COMPLETE

Tiptoe Progress

Slack Channel	#psp-audit-prod-tiptoe
Hourly Monitoring	Tip-Toe Monitoring

Duration

Tiptoe Start Time	Tiptoe End Time	Duration
17 Sept 2022 12PM IST	18 Sept 2022 2PM IST	~24 Hours

Tiptoe Highlights

- **No Functional issues** observed.
- **10K payroll runs** completed, **1850 Vendor Payment runs** completed and no failures observed.
- **648K requests** served without any issues.
- **3407 batch job runs** completed and **21 unique batch jobs** completed successfully.
- **14K employees viewed paychecks** via Workforce Portal.
- **12K+ paycheck messages** were published, and **risk assessed**.
- 46 Reports were generated, and 37 unique report jobs completed.
- **Replication Lag < 10seconds** between Postgres and Oracle, and **data parity looks good**.
- Not observed any issues in Application and Database infrastructure, and the metrics look good.

Downtime Summary

Downtime	Estimated	Actual	Tiptoe/Rollback Window		Comments
			Start Time	End Time	
Tiptoe	120 Mins	110 Mins	12.22 PM (Sep 17)	02:12 PM (Sep 17)	Time Split Expected • 30 Mins of PSP One Pod deployment. • 1 hour of Deploying all apps at full scale. • 30 Mins for database side steps ◦ Database Replication Lag is 0. ◦ Stopping A B and creating B A replication tasks or vice versa in case of rollback.
Planned Rollback	130 Mins	51 Mins	12.44 PM (Sep 18)	01:35 PM (Sep 18)	Time Split Expected • Database Replication Lag is 0. • 10 Mins for DB Solo validation time. (Assuming Sunday 11AM DBSolo Run).
Emergency Rollback (In case of issues during the Tiptoe window)	180 Mins	Not Executed	Not Executed	Not Executed	Time Split Expected • 130 Mins for Planned Rollback • 30 Mins Debugging and Replication Lag to settle down. • 20 Mins of additional time for DBSolo. (DBSolo will take 30 mins here)

Learnings/Enhancements

S No	Learning	Action Item	Owner	Status	Comments
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1	There was a DMS user issue during the replication tasks creation. We were able to resolve with the help of AWS support.	Its one time activity. Oracle(A) was going to be target from Postgres source and its completed. There was no clear documentation that this user will automatically gets created via DMS. AWS confirmed that we can create and start replication.	Choubey, Mayank	IN PROGRESS	- Add pre-cutover checklist step for DMS user in cutover wiki. - check-in user creation script in github
2	We identified 2 workflows performance can be improved		Sur, Koustav Swain, Amrit	IN PROGRESS	
3	Few app startup logs were not appearing in Splunk and one of the validation dashboards was not populating due to this. We were able to validate functionality through other mechanisms. This is one time log, and we are checking the reason.		Bansal, Jatin	IN PROGRESS	
4	Ensure No Batch Jobs are stuck.	To add a step in Cutover.	N, Ramasubramani	OPEN	
5	Create new user tickets just before the Gateway validation for Sanity.	This will be taken care	Everyone.	OPEN	
6	Filling up monitoring metrics is taking time		Everyone	OPEN	
7	Replication tasks source is not checked in to Github	Replication task source was already checked-in at github . Automation was not checked-in as it was created later. Will be taken care.	Choubey, Mayank	IN PROGRESS	
8	There was a issue during Replication task creation	Automation was created later. This is completed	Choubey, Mayank	IN PROGRESS	check-in user creation script in github
9	We are disabling the constraints	Yes, its already enabled with NOVALIDATE state in current Production as well.	Choubey, Mayank	COMPLETED	
10	Some of the database validation queries are not in wiki, check in to Github as well		Choubey, Mayank	IN PROGRESS	- updated wiki and github check-in - Check-in validation queries in github
11	Why 3 users while going from Oracle -> Postgres but only 2 users while going from Postgres -> Oracle ?	We are locking readonly user as well in postgres, can be updated	Choubey, Mayank	COMPLETED	
12	Check in the image validation script		Hoolimath, Praveenkumar	OPEN	
13	Ensure the job count in PSP and RJF (Ensure scheduling & unscheduling)		Agarwal, Akshit	OPEN	
14	Ensure all the unnecessary jobs (BulkInviteProcessor) are not running		Agarwal, Akshit	OPEN	
15	Apps - Parallel Scale up & down.		Hoolimath, Praveenkumar	COMPLETED	No Action needed

16	AWS Support stand by.		Choubey, Mayank	IN PROGRESS	Add support case for DMS and ask to join cutover zoom call
17	Inform interested ppl to join.		N, Ramasubramani	IN PROGRESS	
18	Dev team having DB access.	This will be taken care.	Choubey, Mayank	IN PROGRESS	
19	Enhance DB Replication Task	Completed as part of Step#8 above	Choubey, Mayank	COMPLETED	
20	Resolve Permission issue for AWR user		Choubey, Mayank	IN PROGRESS	
21	Fix WARNING/ERROR "database 1 does not exit"		Choubey, Mayank	IN PROGRESS	
22	Add alert for any DB validation OOS(Out of Sync)		Kumar, Shobhit	OPEN	
23	Cleanup unused replication slots from Postgres db		Choubey, Mayank	OPEN	
24	- Add decommissioning of Oracle (A) and Oracle (C) and corresponding DMS replication resources (wiki) - Duration to keep this running and rational behind it		Choubey, Mayank	OPEN	
25	Decommission exiting UIP pipeline and delete existing data from Oracle audit schema in Datalake		Kumar, Shobhit	OPEN	

Important Resources

Dev Portal

Application	Dev Portal Link
PSP	PSP Dev Portal
PAPI	PAPI Dev Portal
PubSub	Pubsub Dev Portal
RJF	RJF Dev Portal
PayrollIDM	PayrollIDM Dev Portal

JSS

Application	JSS Link
PSP Batch Jobs	PSP JSS View
RJF Reports	RJF JSS View

Splunk

Type	Link
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Application	Splunk Application Production
Gateway	Splunk Gateway Production

ArgoCd

Environment	Application	Dev Portal Link
Production	PSP	Production PSP ArgoCd
	PAPI	Production PAPI ArgoCd
	Pubsub	Production ArgoCd
	RJF	Production RJF ArgoCd
	PayrollDM	Production PayrollDM ArgoCd

Notification Email Addresses

 SBSEG SBG Payroll PSP Informed <SBGPAYROLLPSPInformed@[intuit.com](mailto:SBGPAYROLLPSPInformed@intuit.com)>; SBSEG-PD DTPayrollExtended <SBSEG-PDDTPayrollExtended@[intuit.com](mailto:SBSEG-PDDTPayrollExtended@intuit.com)>;

Change Requests

Application	Tiptoe			Rollback		
	Change Request	Deployment Image	Status	Change Request	Deployment Image	Status
PSP	CHG1813010	master-1043fea-eks	COMPLETED	CHG1814961	master-1043fea-eks	COMPLETED
PAPI	CHG1813051	master-7a7dc83	COMPLETED	CHG1814963	master-7a7dc83	COMPLETED
Pubsub	CHG1813059	master-cd91433	COMPLETED	CHG1814964	master-cd91433	COMPLETED
RJF	CHG1813063	master-871f0b7	COMPLETED	CHG1814966	master-871f0b7	COMPLETED
PayrollDM	CHG1813069	master-9f974ac	COMPLETED	CHG1814968	master-9f974ac	COMPLETED

Runbook

Pre Tiptoe Steps

S No	Task	Steps	Due Date	Status	Owner
1	Partner Sign-off.	Refer NotificationEmailAddresses Section. Partner Contacts : Partner Team	30 Aug 2022	COMPLETED	Nagendra, Subhmita, Nare Pras
2	Inform partner teams slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. #payroll-pd-releases #ioc #psp-rtb-priority	01 Sept 2022	COMPLETED	N, Ram
3	Send Release Freeze Email to Dev Team.	No Releases in PSP/PAPI/Pubsub/RJF/PayrollDM allowed in this interval.	30 Aug 2022	COMPLETED	N, Ram

4	Care Sign-off	SAP can be validated in Staging Environment.	30 Aug 2022	COMPLETED	Sur, Kous													
5	Inform Leadership Team on the Risk & Downtime		01 Sept 2022	COMPLETED	Nasi Syec Iqba													
6	Prepare QBDT Production Setup.	Prepare QBDT Production Setup.	30 Aug 2022	COMPLETED	Hoolh, Pravuma													
7	Time for Validating Latest data during the Tiptoe window	Kumar, Shobhit to update expected numbers.<table><thead><tr><th>Activity</th><th>Expected</th><th>Actual</th></tr></thead><tbody><tr><td rowspan="5">Tiptoe</td><td>2 hours - 1 min</td><td rowspan="5"></td></tr><tr><td>4 hours - 2 min</td></tr><tr><td>6 hours - 3 min</td></tr><tr><td>12 hours - 3 min</td></tr><tr><td>24 hours - 5 mins</td></tr><tr><td>Rollback</td><td></td><td></td></tr></tbody></table>	Activity	Expected	Actual	Tiptoe	2 hours - 1 min		4 hours - 2 min	6 hours - 3 min	12 hours - 3 min	24 hours - 5 mins	Rollback			30 Aug 2022	COMPLETED	Kum Shot
Activity	Expected	Actual																
Tiptoe	2 hours - 1 min																	
	4 hours - 2 min																	
	6 hours - 3 min																	
	12 hours - 3 min																	
	24 hours - 5 mins																	
Rollback																		
8	Decide strategy for Monthly & Quarterly Jobs.	Decide strategy for Monthly & Quarterly Jobs.	30 Aug 2022	COMPLETED	Swa Amri													
9	Disable Canary during Tiptoe.	Disable Canary during Tiptoe to reduce time taken.	30 Aug 2022	COMPLETED	Josh Vishth													
10	Make sure there is no IKS Cluster Upgrade during the Tiptoe window.	- Talk to MSaaS team and confirm. - Get the contacts of MSaaS team in case if we need assistance during tiptoe. - Signoff - IKS Signoff & IKS Cluster Upgrade Calendar.	30 Aug 2022	COMPLETED	N, Ramram													
11	Make sure there is no Splunk Maintenance during the Tiptoe window.	Slack Channel : #pi-tools-splunk-spt Conversation : Splunk Signoff	08 Sept 2022	COMPLETED	N, Ramram													

12	We use PRs for the release. Make sure there is no Git maintenance during the Tiptoe window.	- Talk to Git team and confirm. - Signoff - GIT Signoff	30 Aug 2022	COMPLETED	N, Ram ram																		
13	Capture 03 Sept 2022 - 04 Sept 2022 traffic for each app.	- Update AppsHealth section with the last week traffic. - Post the traffic and health numbers in #psp-audit-prod-tiptoe channel	01 Sept 2022	COMPLETED	Agar Aksh Sur, Kous																		
14	Make sure the last Master branch deployed did not have any deployment errors.	Look for errors in this dashboard <table><thead><tr><th>Application</th><th>Dashboard</th><th>Status</th></tr></thead><tbody><tr><td>PSP</td><td>PSP Unknown Errors</td><td>PASS</td></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td><td>PASS</td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td><td>PASS</td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td><td>PASS</td></tr><tr><td>PayrollDM</td><td>PayrollDM Unknown Errors</td><td>PASS</td></tr></tbody></table>	Application	Dashboard	Status	PSP	PSP Unknown Errors	PASS	PAPI	PAPI Unknown Errors	PASS	Pubsub	PubSub Unknown Errors	PASS	RJF	RJF Unknown Errors	PASS	PayrollDM	PayrollDM Unknown Errors	PASS	30 Aug 2022	COMPLETED	Sur, Kous / Agar, Lat
Application	Dashboard	Status																					
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RJF	RJF Unknown Errors	PASS																					
PayrollDM	PayrollDM Unknown Errors	PASS																					
15	Meet to decide GO/No GO	Email Sent. Meeting scheduled for 30th Aug.	30 Aug 2022	COMPLETED	N, Ram ram uthu am, Kartl anNi: Syeed Iqba ubey May: Kum Shot																		
16	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section. Ask the stake holders to be prepared for the downtime.	30 Aug 2022	COMPLETED	N, Ram ram																		

17	Reassign PagerDuty.	1. Update PagerDuty Schedule for the Tiptoe window. 2. Update PagerDuty Schedule for the next week after the Tiptoe. Pager Duty Schedules QBPY - DPS : EMS OPS RNO Primary SBG - QBPY - DPS : DevOps PSP (IDC)	01 Sept 2022	COMPLETED	N, Ramram
18	Note down all the PSP JSS Jobs in the same window.	Link : PSP JSS View Update Failed/MissedBatchJobs Section.	01 Sept 2022	COMPLETED	Sur, Kous
19	Note down all the RJF JSS Jobs in the same window.	Link : RJF JSS View Update Failed/MissedBatchJobs Section.	01 Sept 2022	COMPLETED	Sur, Kous
20	Update IDPS PostgreSQL Credentials. 1. data Access Secondary - url - pg 2. data Access Secondary - password - pg	• Add Secrets from Command Line. PostgresAddSecret.sh • Validate Secrets from Command Line. PostgresValidateSecret.sh • Validate from Developer Portal	01 Sept 2022	COMPLETED	Bans Jatin
21	Update CR and Image Details	Refer ChangeRequests Section	01 Sept 2022	COMPLETED	N, Ramram
22	Create & Review Zero Pod Deployment PRs for each application.	Update ZeroPodDeploymentPR Section	30 Aug 2022	COMPLETED	Josh Vishth H, Prath, Prath

23	Create & Review One Pod Deployment PRs for each application.	Update OnePodDeploymentPR Section	30 Aug 2022	COMPLETED	Josh Vishth Hathi, Pravuma																		
24	Create Payroll DM Tiptoe /Rollback Config PRs	Update OnePodDeploymentPR Section	07 Sept 2022	COMPLETED	Agar Akshat, Jatin																		
25	Bring down staging and middle ware apps.	• Merge ZeroPodDeploymentPR Section for Middleware and Staging. • Ensure no database connections from Middleware and Staging. • Bringdown the databases (pspparmo, psp-par-aud; it will remain down for 7 days) - Done on 16-Sep-22 4:32 PM IST	01 Sept 2022	COMPLETED	Agar Akshat, Choi Mayank																		
26	Ensure there is no OOS or any data issue between Oracle & Postgres	• db solo reports should not have any oos • ensure latest/last 1 day compare is green. Kumar, Shobhit to update more details here.	17 Sept 2022	COMPLETED	Kumar Shobhit																		
27	Validate Product Audit PostgreSQL Database Connectivity.	Manually Connect to Database with the usernames. • ibobadm_owner • ibob_prod_pspapp	30 Aug 2022	COMPLETED	Choi Mayank																		
28	Run Sanity for PSP /PAPI /Pubsub/RJF. Execute the sanity at 11AM on the Tiptoe day.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><thead><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr></thead><tbody><tr><td>PSP</td><td>PSP Sanity</td><td>COMPLETED</td></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td>COMPLETED</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td>COMPLETED</td></tr><tr><td>RJF</td><td>RJF Sanity</td><td>COMPLETED</td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td>COMPLETED</td></tr></tbody></table>	Application	Sanity Link	Status	PSP	PSP Sanity	COMPLETED	PAPI	PAPI Sanity	COMPLETED	Pubsub	Pubsub Sanity	COMPLETED	RJF	RJF Sanity	COMPLETED	PayrollDM	PayrollDM Sanity	COMPLETED	17 Sept 2022	COMPLETED	Bansal Jatin
Application	Sanity Link	Status																					
PSP	PSP Sanity	COMPLETED																					
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Pubsub	Pubsub Sanity	COMPLETED																					
RJF	RJF Sanity	COMPLETED																					
PayrollDM	PayrollDM Sanity	COMPLETED																					
29	Create Change Requests for Tiptoe and Rollback	Refer ChangeRequests Section.	02 Sept 2022	COMPLETED	N. Ramram																		
30	Validate PSP Node Level Sanity.	Validate Node level Sanity and Dashboard Node Level Sanity Dashboard .	01 Sept 2022	COMPLETED	Bansal Jatin																		

31	Payroll DM Tiptoe connecting to Read Only User	Validate PayrollDM Deployment and ensure DT Payroll Postgres populates Audit Postgres Section.	01 Sept 2022	COMPLETED	Agar Akshat Jatin
32	Update Security Group to add Production Connection	Update Security Group to add Production Connection	01 Sept 2022	COMPLETED	Choi Mayank
33	Simulate Dead lock Scenario in pre-prod	• Simulate Dead Lock Scenario in Preprod. • Add the monitoring queries. • Add the steps to resolve. Attach required scripts here.	01 Sept 2022	COMPLETED	Kumar Shobhit
34	Integration Test Suite	Integration Test Suite Results Dashboard ▸ iks-intg ▸ #1630 ▸ Parameters Back to Project Status Changes Console Output View as plain text Edit Build Information Delete build '#1630' Parameters Timings Git Build Data Nexus IQ Policy Evaluation Build #1630 Parameters BRANCH master Specify the branch name to be built. MONOLITH oracle Select primary database to connect to AUDIT postgres Select secondary database to connect to Dashboard ▸ iks-intg ▸ #1630 View as plain text Edit Build Information Delete build '#1630' Parameters Timings Git Build Data Nexus IQ Policy Evaluation Nexus IQ Build Report Test Result Rebuild Build Support Splunk Open Blue Ocean Restart from Stage Play Pipeline Steps Workspaces Previous Build Next Build This run spent: • 4 ms waiting; • 5 hr 52 min build duration; • 5 hr 52 min total from scheduled to completion. Revision: 1043fa535cbeed0bb6a7ace20b42b8e12b20cd Repository: https://github.com/intuit.com/payroll-dt-payroll/dt-payroll-svc.git • origin/master The following steps that have been detected may have insecure interpolation of sensitive variables (click here for an explanation): • withEnv: [NEXUS_SNAPSHOT_REPO_CREDENTIALS_PSW, NEXUS_RELEASE_REPO_CREDENTIALS_PSW] Nexus IQ Policy Evaluation (view report) IQ Server : BP Nexus IQ Server IQ Server URL : https://iq.security.ctodev.intuit.com/vest/ Application : 1730525749114285769 IQ Stage : build 0 Violations affecting 0 components 0 grandfathered Test Result (10 failures / -) com.intuit.sbd.payroll.psp.adapters.dts.v1.8.PSPQueryFinancialTransactionsTests.testMultipleWithInvalid com.intuit.sbd.payroll.psp.adapters.dts.v1.8.PSPQueryFinancialTransactionsTests.testSingleTransaction com.intuit.sbd.payroll.psp.adapters.dts.v1.8.PSPQueryFinancialTransactionsTests.testMultipleTransactions com.intuit.sbd.payroll.psp.adapters.ews.v1.10.CreateAccountProcessTests.createAccountDiskDeliveryAddEin com.intuit.sbd.payroll.psp.adapters.ews.v1.10.CreateAccountProcessTests.createAccountDiskDelivery com.intuit.sbd.payroll.psp.adapters.ews.v1.10.QueryAccountProcessTests.queryAccountCloudDiskDelivery com.intuit.sbd.payroll.psp.batchjobs.amo.ProcessNewAMOMessagesTests.testValidRoleEmailEntitlementUpdateMessage com.intuit.sbd.payroll.psp.batchjobs.amo.ProcessNewAMOMessagesTests.testValidRoleEmailEntitlementUpdateMessage com.intuit.sbd.payroll.psp.domain.EmployeeBETests.validateEmployeeLookupByNonAuthCriteria com.intuit.sbd.payroll.psp.processes.PaymentMethodTests.manualChangeDoesNotGetAutomaticallyChangedBack	15 Sept 2022	COMPLETED	Sur, Koushik / N, Ram

35	Automation Test Suite	PSP Automation Test Results for build - 25514 Environment : iks2_qal Suite : REGRESSION Total workflows ran : 156 Total workflows passed : 156 Total workflows failed : 0 Click here to see detail reports	15 Sept 2022	COMPLETED	Sur, Kous								
36	DBSolo - Data Compare	Update daily compare status and update validation at 11AM before the tiptoe. <table><tr><th>Date & Time</th><th>Status</th></tr><tr><td>Thursday</td><td>MATCH</td></tr><tr><td>Friday</td><td>MATCH</td></tr><tr><td>Saturday 11AM</td><td>MATCH</td></tr></table>	Date & Time	Status	Thursday	MATCH	Friday	MATCH	Saturday 11AM	MATCH	09 Sept 2022	COMPLETED	Kum Shot arlağ Prav Gou
Date & Time	Status												
Thursday	MATCH												
Friday	MATCH												
Saturday 11AM	MATCH												
37	Run data sanity checks for all audit tables.	Data Sanity Script	08 Sept 2022	COMPLETED	Narlı a, Prav Gou								
38	Prepare Stash Commands to replace Oracle A database credentials with Oracle C database credentials	- Download secret.txt - Add prod pem file path to below command - Add correct path for secret.txt - Run the command stash stash put-secret --api-key-file-path <prod.pem file> --vkm vkm.ps.idps.a.intuit.com --secret-name PSPSecrets/secrets/AWSPRD/dataAccessSecondary-url --secret-value secret.txt secret.txt	16 Sept 2022	COMPLETED	Bans Jatin								
39	Manual Connectivity on Oracle C database	Manual Connectivity on Oracle C database.	15 Sept 2022	COMPLETED	Bans Jatin								
40	PayrollDM Tiptoe on Oracle C database	PayrollDM Tiptoe on Oracle C database.	15 Sept 2022	COMPLETED	Bans Jatin								

TipToe Steps

S No	Task	Steps	Status	Owner	Remarks														
1	DBSolo - Ensure Data Validation on Latest Data is complete.	TODO : Kumar, Shobhit to update here.	COMPLETED	Kumar, Shobhit / Narlagalla, Praveen Goud															
2	Inform #ioc slack channel.	Ask the stake holders to be prepared for the downtime. Inform following Slack Channels. #payroll-pd-releases #ioc #psp-rtb-priority CR, Apps & Downtime details to be shared. Refer ChangeRequests Section.	COMPLETED	N, Ramasubramani															
3	Send Downtime Update to stake holders via Email.	CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Refer NotificationEmailAddresses Section.	COMPLETED	N, Ramasubramani															
4	Ensure there are no critical jobs running.	- Navigate to JSS View Link. - Wait for critical jobs to complete.	COMPLETED	Hoolimath, Praveenkumar															
5	Bring down below apps. 1. PayrollDM 2. Pubsub 3. RJF 4. PAPI	- Use ZeroPodDeploymentPR for applications except PSP. - Wait for POD count to become zero.	COMPLETED	Joshi, Vishwanath															
6	Bring down PSP along with Maintenance Page	- Use ZeroPodDeploymentPR for PSP. - Wait for POD count to become zero.	COMPLETED	Joshi, Vishwanath															
7	Ensure all apps are down.	Hit following URLs for each application. <table><tr><th>App</th><th>Health Check URL</th></tr><tr><td>PSP</td><td>PSP Sanity</td></tr><tr><td>SAP</td><td>SAP</td></tr><tr><td>PAPI</td><td>PAPI Health Check</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td></tr><tr><td>RJF</td><td>RJF Health Check</td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td></tr></table>	App	Health Check URL	PSP	PSP Sanity	SAP	SAP	PAPI	PAPI Health Check	Pubsub	Pubsub Sanity	RJF	RJF Health Check	PayrollDM	PayrollDM Health Check	COMPLETED	Agarwal, Akshit	
App	Health Check URL																		
PSP	PSP Sanity																		
SAP	SAP																		
PAPI	PAPI Health Check																		
Pubsub	Pubsub Sanity																		
RJF	RJF Health Check																		
PayrollDM	PayrollDM Health Check																		
8	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	COMPLETED	Muthurangam, Karthikeyan															
9	Note down All the failed RJF /PSP BatchJobs.	Update Failed/MissedBatchJobs Section.	COMPLETED	Dammur, Venkateshwara															
10	Lock Audit Oracle (A instance) users	Lock Oracle Audit Users : IBOBADM and IBOBAPP users. <pre>alter user IBOBADM account lock; alter user IBOB_PROD_PSPAPP account lock;</pre>	COMPLETED	Choubey, Mayank															

11	Kill sessions (if needed)	- Wait for Inflight requests/Running Batch Jobs to complete - time boxed to 2 mins. (talk to App team and remove for next tiptoe) - There should not be any ACTIVE connections to Oracle Audit Database. (talk to App team and remove for next tiptoe) <pre>select 'exec rdsadmin.rdsadmin_util.kill (' s.sid ',' s. serial# ',' ''PROCESS'' ');' from gv\$session s where USERNAME='&username';</pre>	COMPLETED	Choubey, Mayank	
12	Ensure there are no active connections to the Audit Oracle Database.	No active connections from PSP / PAPI / Pubsub / RJF / PayrollDM. <pre>select username,machine,count(*) from gv\$session where username is not NULL and username not in ('SYS','SYSTEM') and type = 'USER' group by service_name,username,machine;</pre>	COMPLETED	Choubey, Mayank	
13	Disable constraint on (A)	<pre>alter table IBOBADM.PSP_QBDT_REQUEST_INFO disable constraint PSP_QBDT_REQUEST_INFO_FK1;</pre>	COMPLETED	Choubey, Mayank	
14	Check DMS Audit Replication Lag	Wait till Audit Oracle --> Audit Postgres Replication lag to zero.	COMPLETED	Kumar, Shobhit	Oracle- Postgres- Replication- Lag
15	Trigger DB Solo to validate Delta Data.	Trigger DB Solo to validate Delta Data. Dashboard Link : DB Validation Dashboard	COMPLETED	Kumar, Shobhit / Narl agalla, Praveen Goud	
16	Apply Repair Scripts if applicable.	Apply Repair Scripts if applicable. Repair Script File Name : Kumar, Shobhit Repair Script Bucket Name : Kumar, Shobhit	COMPLETED	Kumar, Shobhit / Narl agalla, Praveen Goud	
17	Stop A B replication tasks	Once replication lag is confirmed to be 0, stop all A -> B tasks (and never restart , need to keep it for sometime, can be renamed if need)	COMPLETED	Choubey, Mayank / Narl agalla, Praveen Goud	
18	Create B A replication tasks	Create task from B A Verify source and target end points, table mapping for all 3 tasks	COMPLETED	Choubey, Mayank / Narl agalla, Praveen Goud	
19	Start B A replication tasks	Start B A replication tasks	COMPLETED	Choubey, Mayank / Narl agalla, Praveen Goud	
20	Unlock audit Postgres users in Aurora Postgres	<pre>alter user ibobadm_owner with LOGIN; alter user ibob_prod_ospapp with LOGIN; alter user ibobadm_readonly with LOGIN;</pre>	COMPLETED	Choubey, Mayank	

21	Deploy one pod in every PSP App with Postgres Config.	- Sync OnePodDeploymentPR for PSP with Maintenance on. - Ensure correct image is deployed. Refer ArgoCd section. - Monitor Postgres Audit Connection to validate Audit DB URL is pointing to Postgres Database.	COMPLETED	Joshi, Vishwanath / Bansal, Jatin	./imageDetails.sh prd imageDetails.sh									
22	Ensure there are no errors in the startup logs.	Look for errors in this dashboard Unknown Errors	COMPLETED	Agrawal, Lata										
23	Check the Node level Sanity Logs.	Node Level Sanity Dashboard - Check if all the pods passed sanity tests.	COMPLETED	Bansal, Jatin										
24	Monitor DB Connections.	Ensure connections are originating from the enabled PSP POD. <pre>SELECT datname as db,username as username, client_addr as machine, count(*) FROM pg_stat_activity where username not in ('rdsadmin','postgres') group by db,username,machine;</pre>	COMPLETED	Choubey, Mayank										
25	Monitor Overall database /replication health vi CW dashboard	CW Dashboard	COMPLETED	Choubey, Mayank										
26	Scale up PSP Pods without Maintenance with Postgres Config.	- Use ScaleFullDeploymentPR for PSP with Maintenance Off. 1. Sync prod-pg-tt-scale-full-postgres-maintenance which will bring up all the Pods with Postgres Config and with Maintenance. 2. Once all Pods are up, Merge and Sync prod-pg-tt-scale-full-postgres to remove Maintenance and keeping Postgres Config. - Ensure correct image is deployed. Refer ArgoCd section. - Monitor Postgres Audit Connection to validate Audit DB URL is pointing to Postgres Database.	COMPLETED	Joshi, Vishwanath / Bansal, Jatin	./imageDetails.sh prd imageDetails.sh									
27	Run the PSP Sanity.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><thead><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr></thead><tbody><tr><td>PSP</td><td>PSP Sanity</td><td>SUCCESS</td></tr><tr><td>SAP</td><td>SAP Sanity</td><td>SUCCESS</td></tr></tbody></table>	Application	Sanity Link	Status	PSP	PSP Sanity	SUCCESS	SAP	SAP Sanity	SUCCESS	COMPLETED	Bansal, Jatin	
Application	Sanity Link	Status												
PSP	PSP Sanity	SUCCESS												
SAP	SAP Sanity	SUCCESS												
28	Run replication sanity (count check for all 7 tables)	Data sanity scripts	COMPLETED	Narlagalla, Praveen Goud / Kumar, Shobhit										
29	Update the downtime details section for PSP.	Update DowntimeDetails Section.	COMPLETED	Muthurangam, Karthikeyan										
30	Monitor DB Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	COMPLETED	Choubey, Mayank Popat , Ketan Gunvantrai										
31	Bring up all pods for below services 1. PAPI 2. RJF 3. Pubsub 4. PayrollDM	- Merge ScaleFullDeploymentPR for PAPI/RJF/Pubsub/PayrollDM. - Ensure correct image is deployed. Refer ArgoCd section. - Monitor Postgres Audit Connection to validate Audit DB URL is pointing to Postgres Database.	COMPLETED	Bansal, Jatin	./imageDetails.sh prd imageDetails.sh									

32	Ensure there are no errors in the startup logs.	Monitor and Report unknown errors. <table><tr><th>Application</th><th>Dashboard</th></tr><tr><td>PAPI</td><td>PAPI Unknown Errors</td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td></tr><tr><td>PayrollDM</td><td>Payroll DM Unknown Errors</td></tr></table>	Application	Dashboard	PAPI	PAPI Unknown Errors	Pubsub	PubSub Unknown Errors	RJF	RJF Unknown Errors	PayrollDM	Payroll DM Unknown Errors	COMPLETED	Agrawal, Lata						
Application	Dashboard																			
PAPI	PAPI Unknown Errors																			
Pubsub	PubSub Unknown Errors																			
RJF	RJF Unknown Errors																			
PayrollDM	Payroll DM Unknown Errors																			
33	Run PAPI/Pubsub/RJF /PayrollDM Sanity.	Execute below Sanity Suites. <table><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr><tr><td>PAPI</td><td>PAPI Sanity</td><td>SUCCESS</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td>SUCCESS</td></tr><tr><td>RJF</td><td>RJF Sanity</td><td>SUCCESS</td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td>SUCCESS</td></tr></table>	Application	Sanity Link	Status	PAPI	PAPI Sanity	SUCCESS	Pubsub	Pubsub Sanity	SUCCESS	RJF	RJF Sanity	SUCCESS	PayrollDM	PayrollDM Sanity	SUCCESS	COMPLETED	Sur, Koustav	
Application	Sanity Link	Status																		
PAPI	PAPI Sanity	SUCCESS																		
Pubsub	Pubsub Sanity	SUCCESS																		
RJF	RJF Sanity	SUCCESS																		
PayrollDM	PayrollDM Sanity	SUCCESS																		
34	Ensure we see DB Connections from Pubsub/PAPI /RJF/PayrollDM.	SELECT datname as db,username as username, client_addr as machine, count(*) FROM pg_stat_activity where username not in ('rdsadmin','postgres') group by db,username,machine;	COMPLETED	Choubey, Mayank																
35	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	COMPLETED	Muthurangam, Karthikeyan																
36	Run PSP & RJF failed jobs and missed Jobs.	Refer Failed/MissedBatchJobs Section. Run LedgerBalance Job if Missed.	COMPLETED	Dammur, Venkateshwara																
37	Monitor all the apps and databases.	Refer Monitoring Section.	COMPLETED	Everyone																
38	Ensure replication is running without any issues & monitor the lag	TODO : Kumar, Shobhit to update here	COMPLETED	Kumar, Shobhit																
39	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	COMPLETED	Kaura, Varun																
40	Run Payroll From Production Test company	- Validate SST entries. - Validate SST Timezone. - Validate SAP Timezone.	COMPLETED	Hoolimath, Praveenkumar																
41	Monitor RJF & PSP Batch Jobs.	- Update JSS Jobs Schedule completed jobs status every 2hours. - Update RJF Jobs Schedule completed jobs status every 2hours.	COMPLETED	Dammur, Venkateshwara for PSP Agarwal, Akshit for RJF																

Post TipToe Steps

S No	Task	Steps	Status	Owner	Remarks
1	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	COMPLETED	N, Ramasubramani	
2	Offline Monitoring for all the Apps.	Refer Monitoring Section. Update once in every 2 hours in #psp-audit-prod-tiptoe slack channel.	COMPLETED	All	

3	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	COMPLETED	Hoolimath, Praveenkumar	
4	Review & Close all pending CRs.	Refer ChangeRequests Section.	COMPLETED	N, Ramasubramani	
5	Inform partner teams slack channel.	Inform following Slack Channels. #payroll-pd-releases #ioc #psp-rtb-priority	COMPLETED	N, Ramasubramani	
6	Monitor Critical Batch Jobs.	Monitor Critical Batch Jobs (Required Only for the Cutover) Refer CriticalBatchJobsMonitoring Section.	COMPLETED	Dammur, Venkateshwara	
7	Monitor database Stability, Availability & Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	COMPLETED	Choubey, Mayank (Primary) Popat, Ketan Gunvantrai (Escalation)	
8	Monitor replication from Postgres to Oracle	- Monitor DMS replication lag from Postgres to Oracle - Keep running sanity counts - Ensure db solo validation is green from Postgres to Oracle	COMPLETED	Kumar, Shobhit	
9	Review all Rollback PRs.	- Review all Rollback PRs. - Create PRs for PayrollDM.	COMPLETED	Hoolimath, Praveenkumar Joshi, Vishwanath	

Rollback Decision and Action Plan

Issue Category	Scenario	Action Item
Application	Any of the following issues. - P0/P1 customers workflow impacted. - P0/P1 agents workflow impacted. - P0/P1 Batchjobs impacted. Please refer CriticalBatchJobsMonitoring section.	Rollback and Restore application/service.
	P2/P3 customers/agents/BatchJobs issue.	No Rollback. Fix and Move Forward.
Replication	Replication task is running into error and replication is broken due to following scenarios. Data Issue DMS Bug	Initiate immediate rollback, could be data loss depend upon issue detection and replication lag. Data Issue - Error type, no known solution - Task level error/limitation
Database	Postgres Database Issues	Max 30 mins for troubleshooting.
	Block (Physical/Logical) Corruption in Postgres.	Initiate immediate rollback (revisit this once we move towards touchdown)

Rollback Steps

	Task	Steps	Status	Owner	Remarks
1	Ensure Data comparison is clean, run data-fix if applicable. [Skip this step for Emergency Rollback]	TODO : Kumar, Shobhit to update here.	COMPLETED	Kumar, Shobhit	Postgres-Oracle-Replication-Lag

2	Ensure B A Replication lag is 0. [Skip this step for Emergency Rollback]	TODO : Kumar, Shobhit to update here.	COMPLETED	Kumar, Shobhit / Narlagalla, Praveen Goud															
3	[Optional] Consider going to C , if replication is broken from BA or lag is too high only for B A. [Skip this step for Emergency Rollback]	- Decide based on Replication lag & Data Parity check. - Stop BA all replication tasks	NA																
4	Peer review #1 and #2 [Skip this step for Emergency Rollback]	Decide based on Replication lag & Data Parity check.	COMPLETED	Choubey, Mayank / Kumar, Shobhit															
5	[Optional] Ensure no data in overflow partition of SST table as overflow partitions are part of replication now (same to be added to DB Solo)	Do count of rows in overflow partition of SST table. It should be 0.	NA	Kumar, Shobhit / Narlagalla, Praveen Goud															
6	Note down all the PSP JSS Jobs in the same window.	Link : PSP JSS View Update Failed/MissedBatchJobs Section.	COMPLETED	Sur, Koustav	JSS Jobs														
7	Note down all the RJF JSS Jobs in the same window.	Link : RJF JSS View Update Failed/MissedBatchJobs Section.	COMPLETED	Sur, Koustav	RJF Jobs														
8	Inform slack channel about the downtime.	Ask the stake holders to be prepared fo the downtime. Inform following Slack Channels. #payroll-pd-releases #ioc #psp-rtb-priority CR, Apps & Downtime details to be shared. Refer ChangeRequests Section.	COMPLETED	N, Ramasubramani															
9	Send Downtime Update to stake holders via Email.	CR, Apps & Downtime details to be shared. Refer ChangeRequests Section. Refer NotificationEmailAddresses Section.	COMPLETED	N, Ramasubramani															
10	Ensure there are no critical batch jobs/RJF jobs running.	Refer JSS View Link. Wait for critical jobs to complete.	COMPLETED	Dammur, Venkateshwara															
11	Bring down below pods 1. PayrollDM 2. Pubsub 3. RJF 4. PAPI	- Use ZeroPodDeploymentPR for applications except PSP. - Wait for POD count to become zero.	COMPLETED	Hoolimath, Praveenumar															
12	Bring down PSP along with Maintenance Page	- Use ZeroPodDeploymentPR for PSP. - Wait for POD count to become zero.	COMPLETED	Hoolimath, Praveenumar															
13	Ensure all apps are down.	Hit Following URLs and make sure all apps are down. <table><thead><tr><th>Application</th><th>Health Check</th></tr></thead><tbody><tr><td>PSP</td><td>PSP Sanity</td></tr><tr><td>SAP</td><td>SAP</td></tr><tr><td>PAPI</td><td>PAPI Health Check</td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td></tr><tr><td>RJF</td><td>RJF Health Check</td></tr><tr><td>PayrollDM</td><td>PayrollDM Health Check</td></tr></tbody></table>	Application	Health Check	PSP	PSP Sanity	SAP	SAP	PAPI	PAPI Health Check	Pubsub	Pubsub Sanity	RJF	RJF Health Check	PayrollDM	PayrollDM Health Check	COMPLETED	Agarwal, Akshit	
Application	Health Check																		
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PayrollDM	PayrollDM Health Check																		
14	Update the downtime details section for all the apps.	Update DowntimeDetails Section.	COMPLETED	Muthuramgam, Karthikeyan															
15	Note down All the failed RJF/PSP BatchJobs.	Update Failed/MissedBatchJobs Section.	COMPLETED	Dammur, Venkateshwara															

16	Lock Audit Postgres Users	Lock Postgres Audit Users : IBOBADM and IBOBAPP users. <pre>alter user ibobadm_owner with NOLOGIN; alter user ibob_prod_pspapp with NOLOGIN; alter user ibobadm_readonly with NOLOGIN;</pre>	COMPLETED	Choubey, Mayank	
17	Kill sessions (if needed)	- Wait for Inflight requests/Running Batch Jobs to complete. (remove in next tiptoe after checking with app team) - There should not be any connections to Postgres Audit Database. (remove in next tiptoe after checking with app team) <pre>select pg_terminate_backend(pid) from pg_stat_activity where pid in (select pid FROM pg_stat_activity where username in ('ibob_prod_pspapp', 'ibobadm_owner', 'ibobadm_readonly'));</pre>	COMPLETED	Choubey, Mayank	
18	Ensure there are no active connections to the Audit Postgres Database.	No active connections from PSP / PAPI / Pubsub / RJF / PayrollDM. <pre>SELECT dbname as db,username as username,client_addr as machine, count(*) FROM pg_stat_activity where username not in ('rdsadmin','postgres') group by db,username,machine;</pre>	COMPLETED	Choubey, Mayank	
19	Monitor DMS Audit Replication Lag	Wait till Audit Postgres --> Audit Oracle Replication lag to zero.	COMPLETED	Kumar, Shobhit	Postgres-Oracle-Replication-Lag
20	Trigger DB Solo to validate Delta Data.	Trigger DB Solo to validate Delta Data. Dashboard Link : DB Validation Dashboard	COMPLETED	Kumar, Shobhit	
21	Apply Repair Scripts if applicable.	Apply Repair Scripts if applicable. Repair Script File Name : TODOKumar, Shobhit to update Repair Script Bucket Name : TODOKumar, Shobhit to update	COMPLETED	Kumar, Shobhit	
22	[If going to A] Stop B A replication tasks	Once replication lag is confirmed to be 0, stop all B -> A tasks (and never restart , need to keep it for sometime, can be renamed if need)	COMPLETED	Choubey, Mayank / Narlagalla, Praveen Goud	
23	[If going to C] Stop B C replication tasks	Once replication lag is confirmed to be 0, stop all B -> C tasks (and never restart , need to keep it for sometime, can be renamed if need)	NA	Choubey, Mayank / Narlagalla, Praveen Goud	

24	[If going to A] Create A B CDC only replication tasks	Create CDC Only replication task from A B Verify source and target end points, table mapping for all 3 tasks	COMPLETED	Choubey, Mayank / Narlagalla, Praveen Goud	
25	[If going to A] Start A B replication tasks	Start A B	COMPLETED	Choubey, Mayank / Narlagalla, Praveen Goud	
26	[If going to C] Create C B CDC only replication tasks	Create CDC Only replication task from C B Verify source and target end points, table mapping for all 3 tasks	NA	Choubey, Mayank / Narlagalla, Praveen Goud	
27	[If going to C] Start C B replication tasks	Start C B	NA	Choubey, Mayank / Narlagalla, Praveen Goud	
28	Enabled constraint with NOVALIDATE option on new Oracle source database (A or C)	Enable it on A or C based on the decision. <pre>alter table IBOBADM. PSP_QBDT_REQUEST_INFO enable constraint PSP_QBDT_REQUEST_INFO_FK1 NOVALIDATE;</pre>	COMPLETED	Choubey, Mayank / Narlagalla, Praveen Goud	
29	Unlock Audit Oracle Users	- Unlock A or C based on the decision. - Unlock Oracle Audit Users : IBOBADM and IBOBAPP users. <pre>alter user IBOBADM account unlock; alter user IBOB_PROD_PSPAPP account unlock;</pre>	COMPLETED	Choubey, Mayank	
30	Execute this step if we decide to come back to C. Update IDPS credentials for Oracle.	Update IDPS credentials for Oracle.	COMPLETED	Bansal, Jatin	
31	Deploy one pod in every PSP app.	- Use OnePodDeploymentwithOraclePR for PSP with Maintenance on. - Ensure correct image is deployed. Refer ArgoCd section. - Monitor Postgres Audit Connection to validate Audit DB URL is pointing to Postgres Database.	COMPLETED	Hoolimath, Praveenkumar	<pre>. /imageDetails. sh prd</pre> imageDetails.sh
32	Ensure there are no errors in the startup logs.	Look for errors in this dashboard Unknown Errors	COMPLETED	Agrawal, Lata / Sur, Koustav	

33	Monitor DB Connections on Oracle Audit.	<pre>select username,machine,count(*) from gv\$session where username is not NULL and username not in ('SYS','SYSTEM') and type = 'USER' group by service_name,username, machine;</pre>	COMPLETED	Choubey, Mayank											
34	Scale up PSP pods without Maintenance with Oracle Config.	- Use ScaleFullDeploymentwithOraclePR (MergeandSync) for PSP with Maintenance Off. - Sync prod-pg-tt-scale-full-maintenance-oracle which will bring up all the Pods with Oracle Config and with Maintenance. - Once all Pods are up, Merge and Sync prod-pg-tt-scale-full-oracle to remove Maintenance and keeping Oracle Config. - Ensure correct image is deployed. Refer ArgoCd section. - Monitor Postgres Audit Connection to validate Audit DB URL is pointing to Postgres Database.	COMPLETED	Hoolimath, Praveenkumar	<pre>./imageDetails.sh prd</pre> imageDetails.sh										
35	Run the PSP Sanity.	Execute below Sanity Jobs to ensure jobs working fine as expected. <table><thead><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr></thead><tbody><tr><td>PSP</td><td>PSP Sanity</td><td>SUCCESS</td></tr><tr><td>SAP</td><td>SAP Sanity</td><td>SUCCESS</td></tr></tbody></table>	Application	Sanity Link	Status	PSP	PSP Sanity	SUCCESS	SAP	SAP Sanity	SUCCESS	COMPLETED	Hoolimath, Praveenkumar		
Application	Sanity Link	Status													
PSP	PSP Sanity	SUCCESS													
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36	Monitor DB Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	COMPLETED	Choubey, Mayank / Popat, Ketan Gunvantrai											
37	Bring up all pods for below services - PAPI - RJF - Pubsub - PayrollDM	- Use ScaleFullDeploymentwithOraclePR (MergeandSync) for PAPI/RJF/Pubsub/PayrollDM. - Ensure correct image is deployed. Refer ArgoCd section. - Monitor Postgres Audit Connection to validate Audit DB URL is pointing to Postgres Database.	COMPLETED	Hoolimath, Praveenkumar	<pre>./imageDetails.sh prd</pre> imageDetails.sh										
38	Ensure there are no errors in the startup logs.	Monitor and Report unknown errors. <table><thead><tr><th>Application</th><th>Dashboard</th></tr></thead><tbody><tr><td>PAPI</td><td>PAPI Unknown Errors</td></tr><tr><td>Pubsub</td><td>PubSub Unknown Errors</td></tr><tr><td>RJF</td><td>RJF Unknown Errors</td></tr><tr><td>PayrollDM</td><td>Payroll DM Unknown Errors</td></tr></tbody></table>	Application	Dashboard	PAPI	PAPI Unknown Errors	Pubsub	PubSub Unknown Errors	RJF	RJF Unknown Errors	PayrollDM	Payroll DM Unknown Errors	COMPLETED	Agrawal, Lata	
Application	Dashboard														
PAPI	PAPI Unknown Errors														
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RJF	RJF Unknown Errors														
PayrollDM	Payroll DM Unknown Errors														

39	Run PAPI/Pubsub/RJF/PayrollDM Sanity.	Execute below sanity suites. <table><thead><tr><th>Application</th><th>Sanity Link</th><th>Status</th></tr></thead><tbody><tr><td>PAPI</td><td>PAPI Sanity</td><td> SUCCESS </td></tr><tr><td>Pubsub</td><td>Pubsub Sanity</td><td> SUCCESS </td></tr><tr><td>RJF</td><td>RJF Sanity</td><td> SUCCESS </td></tr><tr><td>PayrollDM</td><td>PayrollDM Sanity</td><td> SUCCESS </td></tr></tbody></table>	Application	Sanity Link	Status	PAPI	PAPI Sanity	SUCCESS	Pubsub	Pubsub Sanity	SUCCESS	RJF	RJF Sanity	SUCCESS	PayrollDM	PayrollDM Sanity	SUCCESS	COMPLETED	Bansal, Jatin	
Application	Sanity Link	Status																		
PAPI	PAPI Sanity	SUCCESS																		
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RJF	RJF Sanity	SUCCESS																		
PayrollDM	PayrollDM Sanity	SUCCESS																		
40	Monitor DB Connections.	<pre>select username,machine,count(*) from gv\$session where username is not NULL and username not in ('SYS','SYSTEM') and type = 'USER' group by service_name,username, machine;</pre>	COMPLETED	Choubey, Mayank																
41	Run PSP & RJF failed jobs and missed Jobs.	Refer Failed/MissedBatchJobs Section. Run LedgerBalance Job if Missed.	COMPLETED	Dammur, Venkates hwara																
42	Monitor all the apps and databases.	Refer Monitoring Section.	COMPLETED	All																
43	Ensure replication is running without any issues & monitor the lag	Monitor Replication Lag.	COMPLETED	Kumar, Shobhit																
44	Run Payroll From Production Test company	• Validate SST entries. • Validate SST Timezone. • Validate SAP Timezone.	COMPLETED	Hoolimat h, Praveenkumar																
45	Enable DB deployment and Batch Jobs Scheduling	Enable DB deployment and Batch Jobs Scheduling. Joshi, Vishwanath to update PR here.	COMPLETED	Joshi, Vishwana th																

Post Rollback Steps

S No	Task	Steps	Status	Owner	Remarks
1	Send Downtime Update to stake holders via Email.	Refer NotificationEmailAddresses Section.	COMPLETED	N, Ramasubramani	
2	Offline Monitoring for all Apps for 2 hours.	Refer Monitoring Section.	COMPLETED	All	
3	Review & Take action on all the pending Alerts.	Pending Alerts - Critical Alerts PD Pending Alerts - RNO PD	COMPLETED	Kaura, Varun	
4	Review & Close all pending CRs.	Refer ChangeRequests Section.	COMPLETED	N, Ramasubramani	
5	Inform partner teams slack channel.	Inform following Slack Channels. #payroll-pd-releases #ioc #psp-rtb-priority	COMPLETED	N, Ramasubramani	
7	Monitor Critical Batch Jobs.	Monitor Critical Batch Jobs (Required Only for the Cutover) Refer CriticalBatchJobsMonitoring Section.	COMPLETED	Dammur, Venkateshwara	
8	Monitor database Stability, Availability & Performance	We will be using combination of Performance Insights, monitoring dashboard and adhoc scripts / slow query log based on need	COMPLETED	Choubey, Mayank (Primary) Popat, Ketan Gunvantrai (Escalation)	

9	Monitor replication from Oracle to Postgres	- Monitor DMS replication lag from Oracle to Postgres - Ensure db solo validation is green from Postgres to Oracle	COMPLETED	Kumar, Shobhit	
10	Remove Commented Hook Jobs for PSP	PSP PR : prod-pg-post-tt-revert RJF : environments/prd-usw2-eks (No Merge required, only sync) This step is delayed and this will be taken care on Monday (19th Sep).	COMPLETED	Joshi, Vishwanath Hoolimath, Praveenkumar	

Monitoring

Apps Health

- Tiptoe Hourly Monitoring captured here [Tip-Toe Monitoring](#).
- Health & Traffic summary shared in [#psp-audit-prod-tiptoe](#) slack channel.

Application / Database / Infrastructure Category	Owner	Health			
		Weekend (3rd Sep)	Tiptoe (17th Sep)	Tiptoe (18th Sep)	Rollback (18th Sep)
QBDT Adapter	Tayal, Ankush	HEALTHY	HEALTHY	HEALTHY	HEALTHY
QBDTWS Adapter	Tayal, Ankush	HEALTHY	HEALTHY	HEALTHY	HEALTHY
DIS Adapter	Tayal, Ankush	HEALTHY	HEALTHY	HEALTHY	HEALTHY
ADE Adapter	Bhagat, Suyash	HEALTHY	HEALTHY	HEALTHY	HEALTHY
CDM Adapter	Bhagat, Suyash	HEALTHY	HEALTHY	HEALTHY	HEALTHY
SAP Adapter	Sur, Koustav	HEALTHY	HEALTHY	HEALTHY	HEALTHY
BRM Adapter	Sengar, Ajay / Agarwal, Akshit	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Critical Batch Jobs	Hoolimath, Praveenkumar	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Batch Jobs during Activity	Dammur, Venkateshwara / Hoolimath, Praveenkumar	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Audit Impacted Batch Jobs	Hoolimath, Praveenkumar	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Audit Impacted Workflows	Sur, Koustav	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Postgres Specific Monitoring	Sur, Koustav	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Infrastructure Monitoring	Bansal, Jatin	HEALTHY	HEALTHY	HEALTHY	HEALTHY
PAPI	Rajoria, Himanshu	HEALTHY	HEALTHY	HEALTHY	HEALTHY
RJF	Agrawal, Lata	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Pubsub	Agarwal, Akshit	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Payroll DM	Agarwal, Akshit	HEALTHY	HEALTHY	HEALTHY	HEALTHY
DB Queries	Bansal, Jatin	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Database	Choubey, Mayank	HEALTHY	HEALTHY	HEALTHY	HEALTHY
DBSolo	Kumar, Shobhit	HEALTHY	HEALTHY	HEALTHY	HEALTHY
ArgoCd	Joshi, Vishwanath	HEALTHY	HEALTHY	HEALTHY	HEALTHY
Unknown Errors	Agrawal, Lata	HEALTHY	HEALTHY	HEALTHY	HEALTHY

PSP

S. No	Dashboard	Link	Comments
1	PSP Batch Jobs 1. Successful Jobs 2. Failed Jobs	PSP Batch Jobs	Update JSS Jobs Schedule completed jobs status every 2hours.
2	PSP JSS View 1. Successful Jobs 2. Failed Jobs	PSP JSS View	
3	PSP Service Metrics 1. Response Time 2. No of Requests	PSP Service Metrics	
4	PSP Adapters 1. Response Codes	PSP Adapters	
5	PSP API Gateway 1. TP Numbers (Check - Not Populating) 2. Response Codes	PSP API Gateway	
6	PSP Unknown Exceptions	PSP Unknown Exceptions	
7	Workforce	Workforce	

PAPI

S. No	Dashboard	Link
1	PAPI Gateway 1. TP Numbers (Check - Not Populating) 2. Response Codes	PAPI Gateway
2	PAPI GraphQL Gateway	PAPI GraphQL Gateway

Pubsub

S. No	Dashboard	Link
1	Pubsub (TODO : To enhance to capture all entities)	Pubsub Entity Publish
2	Pubsub TID Analyzer	Pubsub TID Analyzer

Report Job Framework

S. No	Dashboard	Link	Comments
1	RJF JSS View 1. Successful Jobs 2. Failed Jobs	RJF JSS View Splunk Dashboard : QBDT PSP Report Jobs IKS	Update RJF Jobs Schedule completed jobs status every 2hours.

PayrollDM Monitoring

S. No	Dashboard	Link
1	QBDT PSP DG2.0	QBDT PSP DG2.0

Audit Batch Jobs Monitoring

S. No	Steps	Jobs	Validation Points
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1	Monitor Batch Jobs Reading /Writing to Audit Databases	• QbdtUnprocessedRequestsRetry • EoqSUIAdjustments • EntitlementProcessor	QbdtUnprocessedRequestRetry This job only reads from SST Monitor the job when the below query returns results <pre>-- Find the records to be processed by the job Select /*+PARALLEL(8) */ upr.SOURCE_SYSTEM_TRANSMISSION_ID from PSPADM.PSP_QBDT_UNPROCESSED_REQUEST upr where upr.STATUS in ('Queued') and not exists(Select 'T' From PSPADM.PSP_QBDT_UNPROCESSED_REQUEST upr1 Where upr1.COMPANY_FK = upr.COMPANY_FK and upr1.STATUS in ('Error', 'Processing')) order by upr.COMPANY_FK, upr.CREATED_DATE; -- Validate records created in these 2 tables select /*+PARALLEL(8) */ COMPANY_ID, DESCRIPTION, SOURCE_SYSTEM_TRANSMISSION_SEQ, TO_SOURCE_SYSTEM--, REQUEST_DOCUMENT from ibobadm.PSP_SOURCE_SYSTEM_TRANSMISSION where FROM_SOURCE_SYSTEM = 'QBDT' order by CREATED_DATE desc; select /*+PARALLEL(8) */ * from ibobadm.PSP_QBDT_REQUEST_INFO order by CREATED_DATE desc;</pre> EoqSUIAdjustments Only Reads from SST. This is a manually run job and we are not planning to run it in tiptoe window EntitlementProcessor This job makes calls to ERS and AMO, which are logged into SST. <pre>select /*+PARALLEL(8) */ COMPANY_ID, DESCRIPTION, SOURCE_SYSTEM_TRANSMISSION_SEQ, TO_SOURCE_SYSTEM, CREATED_DATE from ibobadm.PSP_SOURCE_SYSTEM_TRANSMISSION where FROM_SOURCE_SYSTEM = 'PSP' and DESCRIPTION like 'ERS%' order by CREATED_DATE desc; select /*+PARALLEL(8) */ COMPANY_ID, DESCRIPTION, SOURCE_SYSTEM_TRANSMISSION_SEQ, TO_SOURCE_SYSTEM--, REQUEST_DOCUMENT from ibobadm.PSP_SOURCE_SYSTEM_TRANSMISSION where FROM_SOURCE_SYSTEM = 'PSP' and DESCRIPTION like 'AMO%' order by CREATED_DATE desc;</pre>
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BatchJobs in Activity Window

S. No	Jobs	Comments
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1.	- EntityEvent - AMO - EmailGateway - EMSBSToBRMDataSyncProcessor - IAMEmailAddressProcessor - CheckPrint - EntitlementProcessor - QbdtUnprocessedRequestsRetry - WorkersCompProcessor - AssistedUsageDataSyncProcessor	
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Critical Batch Jobs Monitoring

S. No	Steps	Jobs
1	Monitor Critical Batch Jobs.	- Primary - Nightly - ACH Tax - ATF - EFTPS All - Ledger Balance - EntityEvent - AMO - EmailGateway - EMSBSToBRMDataSyncProcessor - IAMEmailAddressProcessor - CheckPrint - EntitlementProcessor - QbdtUnprocessedRequestsRetry - WorkersCompProcessor - AssistedUsageDataSyncProcessor

Postgres Specific Monitoring

S.No	Dashboard	Comments
1	DT Payroll Postgres	
2		
3		

Audit Workflows Monitoring

S. No	Workflow /AppName	Validation Points
1	HCM Monitoring	Swain, Amrit to update company id values. <pre> select * from psp_hcm401k_company_policy where company_id='<value>' select * from psp_hcm401k_company_policy where hcm401k_company_policy_seq='<value>' select * from psp_hcm401k_employee_deduction where hcm401k_company_policy_fk='<value>' </pre>
2	Run Payroll	Hoolimath, Praveenkumar will run from test company.

3	SAP Monitoring	<pre> /*psid: 106563027 - sst_seq: edff3abd-23fb-4b27-b216-71dfe901bee8 for big document (request) in prod*/ select * from psp_source_system_transmission where source_system_transmission_seq = '<source_transmission_seq>'; /*psid: 106924271 Seq: c2b536ee-d01a-44ba-aec3-c358e0ac2eb8*/ select * from psp_source_system_transmission where company_id = '<value>' ORDER BY created_date DESC; </pre>
4	QBDT/QBDTWS Monitoring	<pre> -- Validate records created in these 2 tables select /*+PARALLEL(8) */ COMPANY_ID, DESCRIPTION, SOURCE_SYSTEM_TRANSMISSION_SEQ, TO_SOURCE_SYSTEM--, REQUEST_DOCUMENT from ibobadm.PSP_SOURCE_SYSTEM_TRANSMISSION where FROM_SOURCE_SYSTEM = 'QBDT' order by CREATED_DATE desc; select /*+PARALLEL(8) */ * from ibobadm.PSP_QBDT_REQUEST_INFO order by CREATED_DATE desc; </pre>

DB Queries Monitoring

S. No	Dashboard	Link
1	PSP DB Telemetry	Monitor long running queries. PSP DB Telemetry

Database Monitoring

S.No.	Dashboard	Link
1	Monitor Audit Postgres --> Audit Oracle Replication Lag	
2	Ensure db solo validation is green from Postgres to Oracle	DB Validation Dashboard
3	Monitor Connections and Long Running Queries	
4	Postgres Specific Dashboards	
5	Aurora Postgres database dashboard	Wavefront : psp-prod-ibob Splunk: psp-prod-ibob

Infrastructure Monitoring

Golden Gate Signals

Application	Module	Golden Gate Signals Link	Owner
PSP	QBDT Adapter	QBDT Adapter Golden Signals	Bansal, Jatin
	ADE Adapter	ADE Adapter Golden Signals	Bansal, Jatin
	BRM Adapter	BRM Adapter Golden Signals	Bansal, Jatin
	DIS Adapter	DIS Adapter Golden Signals	Bansal, Jatin

	Failed Jobs		
RJF	Missed Jobs		
	Failed Jobs		

Deployment Pull Requests

- Ensure All PRs are reviewed and latest images updated.
- Comment out Batch Jobs Unscheduling and Rescheduling.
- Comment out DB Updates Task.
- For Manifest model of deployment, use object deletion. Use PRs for the remaining deployments.
- All branches to be rebased with master before the activity

Zero Pod Deployment PR

Application	Tiptoe PR	Operation	Owner	Reviewer	Status
PSP Staging	stg-pg-tt-scale-to-zero-master		Agarwal, Akshit	N, Ramasubramani Muthurangam, Karthikeyan	DONE
Parallel Setup Middleware	prd-pg-tt-scale-to-zero-manifest		Agarwal, Akshit	N, Ramasubramani Muthurangam, Karthikeyan	DONE
PayrollDM	prod-pg-tt-scale-to-zero	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
Pubsub	prd-pg-tt-scale-to-zero	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
RJF	prod-pg-tt-scale-to-zero	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
PAPI	prod-pg-tt-scale-to-zero	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS

PSP	prod-pg-tt-scale-to-zero-maintenance	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
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One Pod Deployment with Postgres PR

Application	Tiptoe PR	Operation	Owner	Reviewer	Status
PSP	prod-pg-tt-scale-to-one-maintenance	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS

Scale Full Deployment with Postgres PR (Merge and Sync)

Application	Tiptoe PR	Operation	Owner	Reviewer	Status
PSP	prod-pg-tt-scale-full-postgres-maintenance	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
	prod-pg-tt-scale-full-postgres	Merge(master) and Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
PAPI	prod-pg-tt-scale-full-postgres	Merge(environments/prd-usw2-eks) and Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
Pubsub	prod-pg-tt-scale-full-postgres	Merge(environments/prd-usw2-eks) and Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
RJF	prod-pg-tt-scale-full	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
PayrollDM	Config PR - prod-pg-tt-scale-full-postgres environments/prd-usw2-eks	Merge(master) Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS

Rollback to Oracle

We will be rolling back to oracle at the end of TipToe (and in case of an emergency rollback).

We will be scaling down to Zero pods as per [ZeroPodDeploymentPR](#)

One Pod Deployment with Oracle PR

Application	Tiptoe PR	Operation	Owner	Reviewer	Status
PSP	prod-pg-tt-scale-to-one-maintenance-oracle	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS

Scale Full Deployment with Oracle PR (Merge and Sync)

All branches to be rebased with master before the activity

Application	Tiptoe PR	Operation	Owner	Reviewer	Status
PSP	prod-pg-tt-scale-full-maintenance-oracle	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
	prod-pg-tt-scale-full-oracle	Merge(master) and Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
PAPI	prod-pg-tt-scale-full-oracle	Merge(environments/prd-usw2-eks) and Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
Pubsub	prod-pg-tt-scale-full-oracle	Merge(environments/prd-usw2-eks) and Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
RJF	prod-pg-tt-scale-full	Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS
PayrollDM	Config PR - prod-pg-tt-scale-full-oracle environments/prd-usw2-eks	Merge(master) Sync	Joshi, Vishwanath Hoolimath, Praveenkumar	N, Ramasubramani Muthurangam, Karthikeyan	IN PROGRESS